

EC1208

Principles of Macro-Economy

Summer 2023 Examination

Paper 1 · 1.5 hours · Section A: answer 2 of 3 (40%) · Section B: answer 1 of 2 (60%)

Module	EC1208 — Principles of Macro-Economy
Session	Summer 2023
Structure	Section A: three short questions (equal weight) — answer any two, together worth 40% of the total. Section B: two long questions with equally-weighted parts — answer any one, worth 60% of the total.
This scheme covers	All three Section A questions AND both Section B questions in full, so you can study every possible combination rather than just the ones you'd pick on the day.

A note on reused content. Several questions on this paper (A1's circular flow, A2's Kuznets quote, and all of Question B2) are identical or near-identical to questions on other EC1208 papers in this series — the model answers here draw on that same verified content, adapted to this paper's exact wording and mark structure.

Section A — Short Questions

Answer 2 of 3 · 40% of total marks · each question worth 100% of its own mark allocation

Question A1

100%

Illustrate and explain using a circular flow diagram, how expenditure, production, and income relate to one another.

FULL MODEL ANSWER

The circular flow diagram shows the economy as two connected loops between households and firms, mediated by two types of markets.

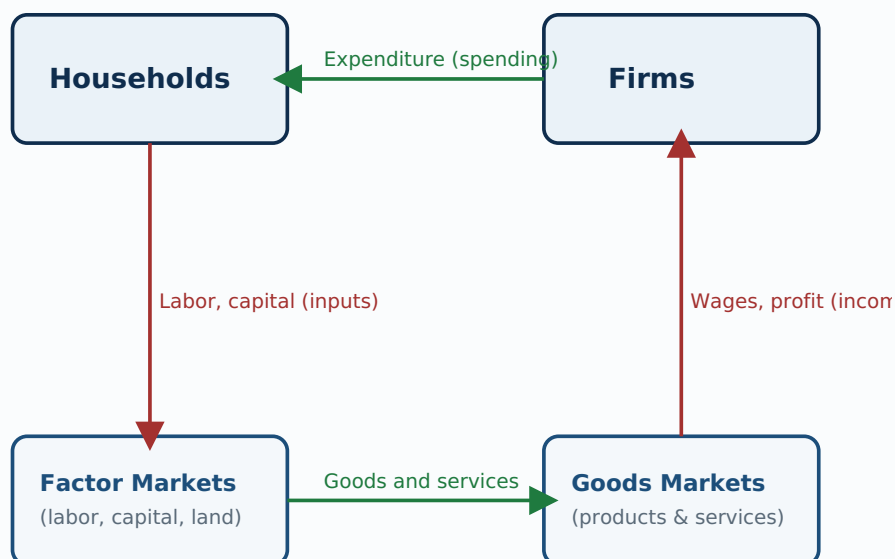


Figure 1: The circular flow diagram. Households supply labor and capital to firms through factor markets, receiving wages and profit (income) in return. Firms supply goods and services to households through goods markets, receiving expenditure (spending) in return. Production, expenditure, and income are three views of the same underlying flow.

The Two Loops

The inner loop — factor markets (red arrows). Households own the economy's factors of production (their own labor, and indirectly, capital through savings and share ownership) and supply them to firms through factor markets. In exchange, firms pay wages, rent, and profit — this is **income** flowing back to households.

The outer loop — goods markets (green arrows). Firms use those factors to **produce** goods and services, which they sell to households through goods markets. Households pay for these goods and services out of their income — this payment is **expenditure**.

Why Production, Expenditure, and Income Must Be Equal

They are three different vantage points on the identical flow of value around the circle. Every euro a firm receives in expenditure from selling its output is, by definition, also the value of what it produced (production), and every euro of that revenue must ultimately be paid out to someone (wages to workers, profit to owners) as income — there is no fourth destination for the money. This is why national income accounting identifies GDP as Production = Expenditure = Income: they are not three independent quantities that happen to be similar, but three equivalent ways of measuring the single flow of value circulating around the diagram.

In practice, GDP is measured using all three approaches as a cross-check. Statistical agencies calculate GDP via the production approach (summing value added across industries), the expenditure approach (summing consumption, investment, government spending, and net exports: $C + I + G + X - M$), and the income approach (summing wages, rents, interest, and profits) — in principle all three yield the same total, and discrepancies between them are used as a check on data quality and to estimate a statistical error term.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Correctly draws/describes both loops (factor markets and goods markets) with households and firms as the two agents, and explicitly explains why production, expenditure, and income must be equal (three views of the same flow), ideally referencing the three GDP measurement approaches.
50-79%	50-79%	Correctly describes the circular flow but with a less complete explanation of why the three measures are necessarily equal.
20-49%	20-49%	Partial or unclear diagram/description; households and firms identified but the flows not clearly connected to production/expenditure/income.
0-19%	0-19%	No meaningful attempt.

Question A2**100%**

Nobel laureate Simon Kuznets, who completed significant work on national income accounts in the 1930s, said that the welfare of a nation can scarcely be inferred from a measurement of national income.

Would you agree with this statement? Why or why not? Explain your answer.

FULL MODEL ANSWER

Agree — and there is a particular irony worth noting in the source of the quote: Simon Kuznets, the economist most responsible for developing the modern national income accounting framework that produces GDP, was himself explicit from the very beginning that his own creation was not a welfare measure. This isn't an external critique of GDP; it's a warning from its own architect.

Agree**Why GDP Cannot Be Read Directly as Welfare**

GDP omits large categories of activity and value entirely. It excludes depreciation of the physical capital stock and natural resources, home production (cooking, cleaning, childcare performed within a household rather than purchased on the market), transactions in the underground or informal economy, and leisure time — despite leisure being something people clearly value and would, in many circumstances, trade income for. A country where citizens work excessive hours with minimal leisure can register a higher GDP than an otherwise identical country with shorter working hours and more leisure, even though the second may plausibly represent higher welfare.

GDP does not net off negative externalities. Pollution, noise, and crime are not subtracted from national income — indeed, the clean-up after an environmental disaster or the private security spending driven by high crime rates adds to GDP (as production of clean-up or security services), even though the underlying events clearly reduce, not increase, genuine welfare.

GDP is silent on distribution. A rising GDP is entirely consistent with rapidly rising inequality, where the gains accrue overwhelmingly to a narrow segment of the population while typical living standards stagnate or fall. The aggregate figure carries no information about how broadly its growth is shared, yet how broadly growth is shared plausibly matters enormously for a nation's overall welfare.

GDP doesn't directly measure subjective wellbeing, health, or security.

The Easterlin Paradox — the finding that reported happiness does not rise in step with a country's rising average income once basic needs are met — is direct empirical evidence that income and welfare/happiness diverge in exactly the way Kuznets's statement warns against. GDP, as a pure measure of production, was never designed to capture happiness, health outcomes, personal security, or other dimensions widely understood to matter for a good life.

The Case for Some Qualification

GDP remains a genuinely useful, objective proxy, even if an imperfect one. GDP correlates strongly with many things people plausibly do value — access to healthcare, nutrition, education, and infrastructure — and it is calculated using a standardised, auditable methodology that permits consistent comparison across countries and over time, something few genuine welfare alternatives can match for regularity and comparability. Kuznets's own point was not that national income is worthless, but specifically that welfare “can scarcely be inferred” from it — a caution about over-interpreting the measure, not a claim that it has no informational value at all.

A strong answer explicitly engages with the irony that Kuznets himself — the person who built the modern national accounting framework — issued this warning, since it lends particular authority to the statement, and distinguishes between GDP being a poor welfare measure versus GDP still being a useful production/activity measure for many policy purposes.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Takes a clear position (agreeing, with reasoned qualification), supports it with multiple specific mechanisms (omitted categories, externalities, distribution, well-being/Easterlin Paradox), and shows critical balance by acknowledging GDP's genuine strengths as an objective, comparable production measure.
50-79%	50-79%	Takes a position and supports it with two or three specific mechanisms, but without the fuller balanced discussion of GDP's genuine strengths.
20-49%	20-49%	General, largely unsupported agreement or disagreement with limited specific reasoning.
0-19%	0-19%	No meaningful attempt, or a fundamental misunderstanding of what GDP measures.

Question A3**100%**

As China's economic growth depends on net exports to a large extent, some economists suggest that China needs to increase its domestic consumption rather than export to maintain its economic growth. However, in the long run, increases in consumption may lower the income. Explain how an increase in consumption may create two contradictory effects (i.e. higher and lower income) on economic growth.

FULL MODEL ANSWER

This apparent contradiction resolves once the time horizon is made explicit: an increase in consumption has one effect operating through short-run aggregate demand, and an entirely different, opposite-signed effect operating through long-run capital accumulation.

Effect 1 (Short Run): Higher Consumption Raises Income

The Keynesian demand-side channel. Consumption (C) is a direct component of GDP: $Y = C + I + G + X - M$. An increase in consumption spending directly and immediately raises aggregate expenditure, and therefore GDP, in the short run. This effect is further amplified by the multiplier: the recipients of that new consumption spending (retailers, manufacturers, their employees) themselves spend a portion of their increased income, generating further rounds of spending and income throughout the economy.

This is the specific argument behind rebalancing China's growth model toward consumption. China's growth has historically leaned heavily on net exports and investment rather than household consumption; economists advocating a shift toward domestic consumption are making exactly this short-run, demand-side argument — that higher consumption spending would directly boost current aggregate demand and GDP, and would also make growth less dependent on volatile external demand for Chinese exports.

Effect 2 (Long Run): Higher Consumption Lowers Income

The Solow growth model's saving-investment channel. In national income accounting, saving is simply the portion of income not consumed: $S = Y$ minus C . Holding income fixed in any given period, an increase in consumption necessarily means a fall in saving. In a closed (or largely closed) economy, saving is the source of funds for investment ($S = I$ in equilibrium) — so lower saving means lower investment in new physical capital.

Lower investment means slower capital accumulation, and eventually a lower level of GDP. The aggregate production function shows that GDP depends positively on the physical capital stock. If higher current consumption crowds out saving and investment, the capital stock accumulates more slowly over time (net of depreciation), meaning the economy approaches a lower steady-state level of capital per worker, and therefore a lower long-run level of GDP, than it would have reached with a higher saving rate. This is precisely the mechanism the Solow growth model uses to link a country's saving rate to its long-run income level.

Reconciling the Two Effects

The contradiction is really a trade-off between short-run demand stimulus and long-run supply capacity. Higher consumption today raises GDP through the demand-side multiplier channel in the short run, but by reducing saving and investment, it slows the accumulation of productive capacity that drives GDP growth over the long run. This is essentially the mirror image of the well-known "paradox of thrift": just as more saving can reduce short-run demand and income (the paradox of thrift) while raising long-run income (via more investment), more consumption raises short-run income (via demand) while potentially reducing long-run income (via less investment) — the same underlying tension between C and S , viewed from the opposite direction.

Applied to China specifically. The policy tension is genuine: shifting toward a more consumption-driven growth model can support GDP in the near term and reduce China's exposure to external demand shocks, but if it comes at the cost of a significantly lower national saving rate, it also risks slowing the capital accumulation that has underpinned China's investment-led growth model for decades — which is exactly why the rebalancing debate involves a genuine economic trade-off, not simply an unambiguous improvement.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Correctly explains BOTH the short-run Keynesian demand channel (C directly raises Y, amplified by the multiplier) AND the long-run saving/investment channel (higher C means lower S, lower I, slower capital accumulation, lower long-run Y), and explicitly reconciles the two as a short-run vs. long-run trade-off, ideally connecting to China's specific rebalancing debate.
50-79%	50-79%	Correctly explains both channels but without clearly reconciling them as operating over different time horizons.
20-49%	20-49%	Explains one channel (typically the short-run demand effect) well but misses or garbles the long-run saving/investment channel.
0-19%	0-19%	No meaningful attempt, or no engagement with why the two effects point in opposite directions.

Section B — Long Questions

Answer 1 of 2 · 60% of total marks · all parts equal weight

Question B1

Productivity, Growth Theory & Institutions

B1 (a)

25%

(a) Explain the term productivity? Why does it vary across countries?

FULL MODEL ANSWER

Productivity refers to how efficiently an economy converts its inputs — physical capital and labor — into output (GDP). Formally, it is captured by the technology index, A , in the aggregate production function $Y = AF(K,H)$: for any given quantities of capital and efficiency units of labor, a higher A means more GDP is produced from the same inputs.

Why Productivity Varies Across Countries

1. Human capital. The stock of skills embodied in a country's workforce — education, training, and accumulated experience — directly affects how much output a given number of workers can produce. Countries with stronger education systems and more accumulated workforce experience have higher effective labor input ($H = L \times h$) even holding the raw number of workers constant.

2. Physical capital. The stock of machinery, equipment, and business infrastructure available to workers varies enormously across countries. More and better capital gives workers more tools to work with, directly raising output per worker — though, as the aggregate production function's diminishing marginal product property shows, this effect diminishes as the capital stock grows relative to the workforce.

3. Technology. Beyond the quantities of capital and labor themselves, countries differ in the superior knowledge and efficiency of their production processes — represented by the technology index A itself. A country with more advanced technology produces more output from identical quantities of capital and labor than a country with less advanced technology; this is the specific channel through which innovation, know-how, and managerial practice affect productivity independent of factor accumulation.

A deeper explanation: institutions shape all three proximate sources.

Beneath these three proximate sources lies a deeper explanation, discussed further in part (d): a country's institutions — property rights, rule of law, political stability — shape the incentives to accumulate human capital (why invest in education if returns aren't secure?), physical capital (why invest if property rights are weak?), and technology (why innovate if the gains can be expropriated?) in the first place. Institutions are therefore often described as the fundamental cause of productivity differences, with human capital, physical capital, and technology as the proximate (more immediately visible) causes.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly defines productivity with reference to the aggregate production function (the role of A), correctly identifies all three proximate sources (human capital, physical capital, technology) with explanation of each, and connects to the deeper institutional explanation.
12-19%	12-19%	Correctly identifies the three proximate sources with reasonable explanation, but without the deeper institutional connection.
5-11%	5-11%	Vague definition of productivity, or only one or two sources identified.
0-4%	0-4%	No meaningful attempt.

B1 (b)**25%**

(b) Illustrate the Solow growth model to answer the following questions.

- i. What is the impact of a higher saving rate on GDP? Explain the process.
- ii. What is the impact of an increase in labor force on GDP? Explain the process.

FULL MODEL ANSWER

The Solow model tracks capital per worker (k) on the horizontal axis against output and investment per worker on the vertical axis. Output per worker, $y = f(k)$, is the familiar concave production function; actual investment per worker is $sf(k)$, a constant fraction s of output; and break-even investment, $(n+d)k$, is the amount of investment needed just to maintain the existing capital-per-worker ratio given population growth (n) and depreciation (d). The steady state, k^* , occurs where these two lines intersect — capital per worker (and therefore output per worker) is stable, neither rising nor falling.

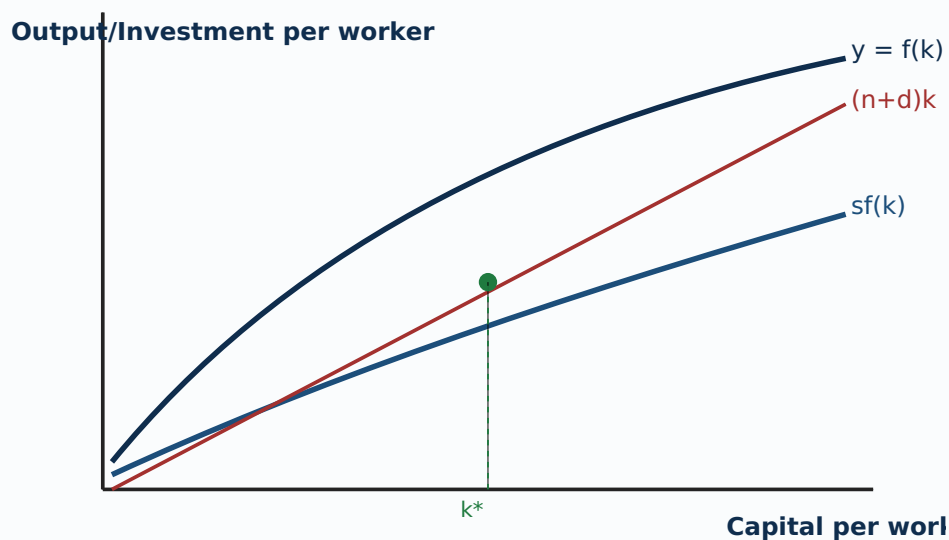


Figure 1: The Solow growth model. Steady-state capital per worker (k^*) occurs where investment, $sf(k)$, exactly equals break-even investment, $(n+d)k$ — the amount needed to replace depreciating capital (d) and equip new workers (n) at the existing capital-per-worker ratio.

(i) Impact of a Higher Saving Rate on GDP

The process: a higher saving rate shifts the investment curve upward.

An increase in the saving rate, s , shifts the $sf(k)$ curve upward at every level of k (since a larger fraction of output is now invested rather than consumed). At the original steady state k^* , actual investment now exceeds break-even investment, so capital per worker begins to rise.

The economy transitions to a new, higher steady state. As k rises, output per worker $y = f(k)$ rises with it, though at a diminishing rate (the concave production function). Capital per worker keeps rising until it reaches a new steady state, k^{**} , at a higher level than before, where the new (higher) $sf(k)$ curve again intersects the unchanged $(n+d)k$ line. GDP per worker is therefore permanently higher in the new steady state — but importantly, this is a level effect, not a permanent growth-rate effect: once the new steady state is reached, GDP per worker stops growing again (absent further increases in s or improvements in technology), since diminishing returns to capital mean saving alone cannot sustain perpetual growth.

(ii) Impact of an Increase in the Labor Force on GDP

The process: population/labor force growth raises the break-even investment line. A higher rate of labor force growth, n , increases the $(n+d)k$ line's slope — more new investment is now needed just to equip the growing number of new workers with the existing amount of capital per worker, on top of replacing depreciation. At the original steady state, break-even investment now exceeds actual investment $sf(k)$, so capital per worker begins to fall.

The steady state shifts to a lower level of capital (and output) per worker. Capital per worker falls until a new, lower steady state is reached, where the steeper $(n+d)k$ line intersects the unchanged $sf(k)$ curve. GDP per worker is therefore lower in the new steady state than before — even though total GDP (not per worker) may still rise, since there are now more workers overall, each producing less than before due to more of them sharing the same capital stock (a lower capital-to-labor ratio).

This distinguishes total GDP growth from GDP per worker/capita growth. A larger labor force mechanically raises total output (more workers producing something), but the model shows that rapid population growth can actually reduce living standards (GDP per worker), since capital has to be spread more thinly — a key reason cross-country comparisons of prosperity focus on GDP per capita rather than total GDP.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly draws/describes the Solow diagram with all three curves labelled, correctly explains the saving rate increase as an upward shift in $sf(k)$ leading to a higher steady state (with the level-vs-growth-rate distinction noted), and correctly explains labor force growth as steepening $(n+d)k$ leading to a lower steady-state capital/output per worker (distinguishing total GDP from GDP per worker).
12-19%	12-19%	Correctly identifies the direction of both effects but with less precise explanation of which curve shifts and why.
5-11%	5-11%	One of the two sub-questions answered correctly; the other vague or missing.
0-4%	0-4%	No meaningful attempt, or no reference to the Solow model's actual structure.

B1 (c)**25%**

(c) What did Malthus predict about economic growth? Did his predictions come true? Why or why not?

FULL MODEL ANSWER

Malthus's Prediction

Humankind was destined to remain at subsistence level, indefinitely.

Writing in 1798, Thomas Malthus argued that any rise in income per capita above the minimum needed for survival (subsistence level) would trigger higher fertility rates. The resulting population growth would then outstrip the (largely fixed, technologically static) capacity of land to produce food, driving income per capita back down to subsistence. Under this logic, sustained improvement in living standards was essentially impossible — any gains would always be reabsorbed by population growth, trapping humanity in a repeating cycle now known as the Malthusian Trap.

Did His Predictions Come True?

True for most of history, but decisively wrong from the eighteenth century onward

The prediction held remarkably well for the vast majority of human history before Malthus wrote. For roughly 2,800 years before 1800, global GDP per capita showed essentially no sustained growth, consistent with economies cycling between temporary improvements and population-driven reversals exactly as Malthus described — strong historical support for the theory as a description of the pre-modern world.

But the prediction failed dramatically from around the mid-eighteenth century onward. Sustained, compounding growth in income per capita began in Britain and then spread globally, with income per capita reaching roughly 12 times its 1750 level by the year 2000 — the exact opposite of a world permanently trapped at subsistence.

Why the Predictions Ultimately Failed

Malthus's model implicitly assumed technology and fertility behaviour would stay fixed. The Malthusian cycle only holds if food-producing technology stays essentially static (so population growth genuinely does outrun the food supply) and if fertility keeps rising whenever income rises above subsistence. Two developments broke both assumptions.

First, the Industrial Revolution created sustained, self-reinforcing technological progress. New machines and production methods, beginning in textile manufacturing and spreading through the economy, allowed output to grow faster than population for the first time in a durable, ongoing way — unlike earlier isolated bursts of innovation, each advance enabled further advances rather than petering out.

Second, the demographic transition broke the assumed fertility response to rising income. As populations urbanised and moved from agriculture into manufacturing, fertility rates fell rather than rose as incomes increased — children shifted from productive farm labour to a costly, education-intensive investment, reducing the incentive for large families. This is precisely the opposite of what Malthus's model assumed would happen as income rose.

Together, these two changes let rising income actually persist, rather than being reabsorbed by population growth. It's the *combination* of technological progress and falling fertility that explains the escape from the Malthusian Trap — technological progress alone, without the accompanying fertility change, would likely have simply been absorbed by a larger population, exactly as it always had been in every earlier episode of innovation before the eighteenth century.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly explains the Malthusian Trap mechanism, correctly notes the prediction held for most of pre-1800 history but failed thereafter, and explains both reasons for the failure (Industrial Revolution technology AND the demographic transition), with the interaction between them made explicit.
12-19%	12-19%	Correctly explains the Malthusian mechanism and identifies the Industrial Revolution as the reason it failed, but without the demographic transition/fertility dimension.
5-11%	5-11%	Vague or partial description of Malthus's prediction without a clear explanation of why it failed.
0-4%	0-4%	No meaningful attempt.

B1 (d)**25%**

(d) As defined by economic historian Douglass North, (1990) “institutions are the rules of the game in a society or, more formally, are the humanly devised constraints that shape human interaction.”

i. Does the role of institutions or geography offer a more plausible explanation of cross-country differences in economic growth? Explain with reference to the work of Daron Acemoglu.

ii. How does the institution hypothesis explain the difference, in terms of economic growth, between North and South Korea?

FULL MODEL ANSWER**(i) Institutions vs. Geography****Institutions offer the more plausible explanation**

The geography hypothesis, and its limitations. Geographic theories attribute cross-country income differences to climate, natural resources, disease burden, and access to trade routes — tropical regions, for instance, have historically faced worse agricultural productivity and higher disease burdens than temperate regions, which some scholars (such as Jeffrey Sachs) argue plays a first-order role in explaining persistent poverty. Geography clearly matters at the margin, but it struggles to explain cases where genetically, climatically, and geographically near-identical populations end up with vastly different economic outcomes purely because of a border.

Acemoglu's reversal-of-fortune evidence directly challenges the geography hypothesis. Daron Acemoglu, alongside Simon Johnson and James Robinson, documented that regions which were relatively prosperous 500 years ago (before European colonisation) are frequently relatively poor today, and vice versa — a “reversal of fortune” that is very difficult to explain through geography alone, since a region's climate and natural endowments haven't changed over that period, but its relative prosperity has reversed. Acemoglu and Robinson argue the explanation lies in the type of institutions European colonisers established: in regions with a high disease burden for European settlers, colonisers tended to install extractive institutions (designed to extract resources with minimal investment in long-run development); in regions more hospitable to European settlement, colonisers built inclusive institutions resembling those of their home countries (secure property rights, broadly distributed political power) — and it is this institutional difference, not the underlying geography, that best predicts subsequent economic performance.

Twin-city and divided-country natural experiments make the same point even more directly. Comparisons like Nogales (a single city split by the US-Mexico border) or North versus South Korea (below) hold geography, climate, and ancestry constant while institutions diverge sharply — and in every such case, it is the institutional divergence, not any geographic difference (since there is none), that tracks the resulting gap in prosperity.

(ii) North Korea vs. South Korea

A near-perfect natural experiment for isolating institutions from every other factor. Before their division following the Second World War, the Korean peninsula was a single people sharing identical geography, climate, culture, language, and history. Since 1948, the two halves have followed dramatically different institutional paths: South Korea developed increasingly inclusive economic and political institutions (secure property rights, market-based resource allocation, and eventually democratic political institutions), while North Korea adopted a centrally planned, extractive economic system under an authoritarian political regime with no meaningful checks on power.

The economic outcomes track the institutional divergence almost exactly. South Korea transformed from one of the poorest countries in the world in the 1950s into a high-income, technologically advanced economy, while North Korea has remained impoverished, with satellite imagery of the peninsula at night showing North Korea almost entirely dark compared to a brightly lit South — a visually stark illustration of the underlying economic gap. Because every other plausible explanatory factor (geography, culture, initial history) is held constant between the two, the institution hypothesis identifies this divergent economic performance as attributable specifically to the different economic and political institutions each half subsequently adopted, not to anything that predates the division.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Takes a clear position favouring institutions, correctly explains Acemoglu's reversal-of-fortune evidence and the extractive-vs-inclusive institutions framework, AND correctly explains why North/South Korea is such a clean test of the institution hypothesis (geography/culture/history held constant, institutions vary, outcomes track institutions).
12-19%	12-19%	Correctly favours institutions with a reasonable explanation, but without the specific Acemoglu reversal-of-fortune evidence or with a less complete Korea explanation.
5-11%	5-11%	Vague preference for institutions or geography without engaging with the Acemoglu literature or the Korea comparison specifically.
0-4%	0-4%	No meaningful attempt.

Question B1 Total: 100% (60% of the Paper)

Question B2

Business Cycles, Multipliers & Stabilisation Policy

B2 (a)**25%**

(a) What are the major sources of fluctuations identified by each of these three schools of thought (i) real business cycle theory (ii) Keynesian theory, and (iii) financial and monetary theories?

FULL MODEL ANSWER

Macroeconomists have identified three broad schools of thought explaining what actually drives short-run economic fluctuations, each emphasising a different underlying source of shocks.

(i) Real Business Cycle Theory

Source of fluctuations: productivity and technology shocks. Real business cycle theory attributes fluctuations to changes in productivity and technology. A wave of productivity-enhancing innovation causes an economic expansion, as firms can produce more output from the same inputs. Conversely, a rise in the price of a key input — oil being the classic example — acts as a negative supply shock and can trigger a recession, since it raises production costs economy-wide independent of demand conditions.

(ii) Keynesian Theory

Source of fluctuations: shifting expectations and "animal spirits." Keynesian theory focuses on changes in business and consumer expectations about the future. "Animal spirits" — John Maynard Keynes's term for the psychological factors driving shifts in business and consumer mood or sentiment — can independently trigger either decreases in spending (recessions) or increases in spending (expansions), regardless of whether any change in underlying productivity or fundamentals has actually occurred. A negative shock can generate pessimism; willingness to spend falls, and — critically for the Keynesian mechanism — that initial fall in spending is not offset by increased spending elsewhere in the economy, and is instead amplified further through multiplier effects as one person's reduced spending becomes another person's reduced income.

(iii) Financial and Monetary Theory

Source of fluctuations: changes in the money supply, prices, and interest rates. Associated principally with Milton Friedman, this view attributes fluctuations to monetary factors: a decrease in the money supply (M2) causes the price level to fall, which reduces employment because wages are downwardly rigid (they don't easily adjust down to match falling prices, so real labour costs effectively rise, and firms cut jobs). A falling money supply also raises the real interest rate, which reduces investment spending by firms, compounding the contractionary effect.

A strong answer keeps the three theories genuinely distinct rather than blending them — RBC theory locates the shock in the real, productive side of the economy (technology/productivity); Keynesian theory locates it in psychology and expectations (demand-side, amplified by multipliers); and financial/monetary theory locates it in the money supply and interest rates. Real-world recessions are often explained using elements of more than one theory simultaneously, which is a reasonable point to make, but the theories themselves are analytically separate.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly identifies the distinct source of fluctuations for all three schools (RBC: productivity/technology shocks; Keynesian: animal spirits/expectations amplified by multipliers; financial/monetary: money supply, price level, and interest rates), with the specific mechanism explained for each rather than just named.
12-19	12-19	Correctly identifies two of the three theories with reasonable mechanism detail, the third vague or missing.
5-11	5-11	Names the three schools but without correctly distinguishing their underlying mechanisms.
0-4	0-4	No meaningful attempt.

B2 (b)**25%**

(b) The concept of multipliers was one of the key elements of John Maynard Keynes's theory of fluctuations. What is a multiplier? Explain with reference to the recent Páirc Uí Chaoimh development case study used in class.

FULL MODEL ANSWER

A **multiplier** is the mechanism by which an initial change in spending generates a larger total change in economy-wide income and output, because one person's spending becomes another person's income, which is then partly re-spent, generating further rounds of income and spending down the chain.

The Mechanism

Spending circulates and compounds, rather than stopping after one transaction. When one euro of new spending occurs, the recipient of that euro (a worker, a supplier, a landlord) doesn't save it entirely — they spend a portion of it themselves, which becomes income for someone else, who in turn spends a portion of that, and so on. The total increase in economy-wide income ends up larger than the initial injection of spending, with the exact size of the multiplier depending on how much of each round is re-spent versus saved or spent on imports (which leak out of the domestic spending chain).

Multipliers work in both directions, and are central to Keynesian theory precisely because they amplify shocks. An initial fall in spending (driven, say, by a negative shift in animal spirits) doesn't just directly reduce the income of whoever was going to receive that spending — it reduces their spending too, which reduces someone else's income, and so on, amplifying the original shock into a much larger overall contraction. This amplification is exactly why Keynesian theory treats multipliers as one of the central elements explaining how economic fluctuations can become so much larger than the shocks that triggered them.

A Worked Example: The Páirc Uí Chaoimh Redevelopment

The initial injection. Consider the redevelopment of Páirc Uí Chaoimh, the GAA stadium in Cork. The construction spend itself is the initial injection — wages paid to construction workers, payments to building material suppliers, and fees paid to architects and contractors.

The first round of re-spending. The construction workers who receive those wages don't save all of it — they spend a good share of it locally: in Cork city cafes, shops, and pubs near the site, on transport, and on other everyday goods and services. That spending becomes income for the owners and staff of those local businesses.

Further rounds, and the eventual total effect. Those café and shop owners, in turn, spend part of their increased income on their own suppliers, staff wages, and purchases — and so the chain continues, each round smaller than the last as some income is saved or spent outside the local area (a “leakage”) rather than re-spent locally. The eventual total increase in economic activity generated across Cork city, once every round of this chain is added up, is larger than the construction spend alone — this total effect (original spend plus all the induced rounds of re-spending) is the multiplier effect, and is exactly the kind of local economic impact argument regularly used to justify major stadium and infrastructure redevelopments.

A complete answer explains *why* the chain eventually stops (leakages to saving, taxation, and imports mean each round is progressively smaller, so the total effect converges to a finite multiple of the initial spend rather than growing without limit) rather than simply asserting spending “multiplies” without explaining the mechanism or its limit.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly defines the multiplier mechanism (spending becomes income, partly re-spent, generating further rounds), explains why the effect eventually diminishes and converges (leakages to saving/imports), and applies it in a fully worked, specific example (such as Páirc Uí Chaoimh) tracing at least two rounds of the chain.
12-19	12-19	Correctly defines the multiplier concept with a relevant example, but without tracing multiple rounds or explaining why the effect eventually diminishes.
5-11	5-11	Vague reference to spending having "knock-on effects" without a clear mechanism or worked example.
0-4	0-4	No meaningful attempt.

B2 (c)**25%**

(c) What do countercyclical policies mean? During an economic boom, what countercyclical fiscal and monetary policies are used? Please use a diagram of the business cycle to help explain your answer.

FULL MODEL ANSWER

Countercyclical policies are fiscal and monetary policy actions deliberately taken to move *against* the current direction of the business cycle — restraining the economy during a boom (when it is running above trend, as in Figure 2) and stimulating it during a recession (when it is running below trend) — in order to smooth out fluctuations and keep actual GDP closer to its trend path, rather than reinforcing whichever direction it happens to be moving in already.

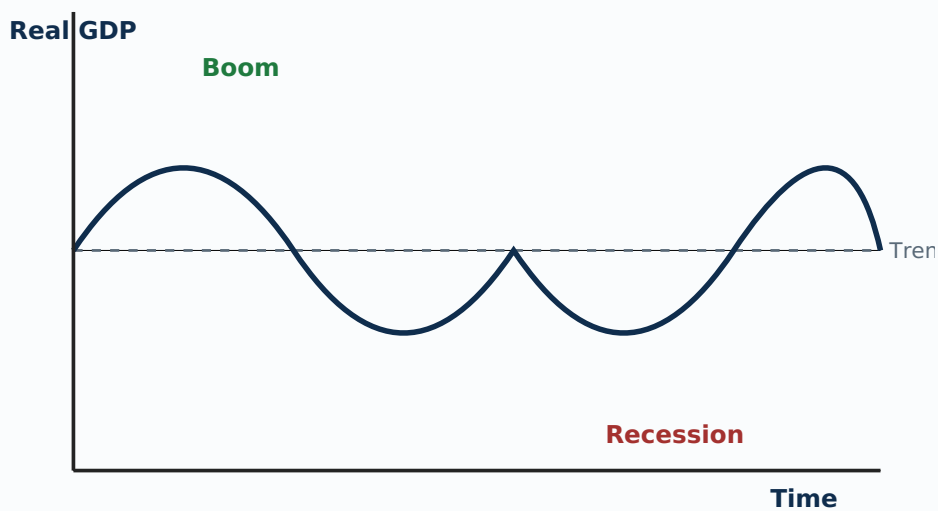


Figure 2: The business cycle — actual real GDP (solid line) fluctuating around its long-run trend (dashed line). A boom is a period where GDP rises above trend; a recession is a period where it falls below trend.

Countercyclical Policy During a Boom

The logic: cool down an economy that is overheating. During a boom, GDP is running above its sustainable trend, which risks fuelling inflation, asset bubbles, and unsustainable borrowing. Countercyclical policy during a boom aims to withdraw some stimulus from the economy specifically to prevent these excesses, rather than letting the boom run unchecked toward a sharper eventual correction.

Countercyclical Fiscal Policy in a Boom

Reduce government spending and/or raise taxes. A government pursuing countercyclical fiscal policy during a boom would restrain its own spending growth and/or raise taxes, withdrawing some private demand from an already-strong economy — the opposite of the spending increases/tax cuts used to fight a recession. In practice this also means resisting the temptation to spend windfall tax revenues (which rise automatically during a boom as incomes and profits rise) and instead saving them, for instance into a sovereign wealth or rainy-day fund, for use during the next downturn.

Countercyclical Monetary Policy in a Boom

Raise interest rates. A central bank pursuing countercyclical monetary policy during a boom raises interest rates, which increases the cost of borrowing for both households and firms, cooling consumption and investment spending, and helping contain inflationary pressure building up from the strong economy — the reverse of the interest rate cuts (and, in more extreme downturns, quantitative easing) used to stimulate a recession.

A common error is describing expansionary policies (spending increases, tax cuts, rate cuts) when asked specifically about boom-time countercyclical policy — those are the correct countercyclical response to a *recession*, precisely the opposite direction to what's being asked here. Countercyclical, by definition, means pushing against the current phase of the cycle: contractionary policy in a boom, expansionary policy in a recession.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly defines countercyclical policy as moving against the current cycle phase, correctly identifies contractionary fiscal policy (spending restraint/tax increases) AND contractionary monetary policy (interest rate rises) as the boom-time response, and uses a correctly labelled business cycle diagram distinguishing boom from recession relative to trend.
12-19	12-19	Correctly identifies contractionary fiscal and monetary policy for a boom but without a clear diagram, or with one of the two policy types missing.
5-11	5-11	Describes expansionary (recession-fighting) policies instead of the boom-time contractionary policies actually asked for.
0-4	0-4	No meaningful attempt.

B2 (d)**25%**

(d) Critically evaluate the fiscal and monetary policies of the Irish government since the turn of the 21st century.

FULL MODEL ANSWER

Since Ireland adopted the euro in 1999, its own government has had no independent monetary policy lever at all — interest rates are set for the whole eurozone by the European Central Bank (ECB). This answer therefore evaluates Irish fiscal policy (which Ireland fully controls) across four distinct periods, and evaluates monetary policy specifically in terms of how well a single ECB rate has fitted Ireland's own economic conditions at each stage.

2000–2008: The Celtic Tiger Boom — Pro-Cyclical Fiscal Policy

Fiscal policy reinforced the boom rather than restraining it. Rather than pursuing countercyclical policy (as discussed in part (c)) by restraining spending and banking windfall revenue during the boom, Irish fiscal policy through the 2000s was broadly **pro-cyclical**: government spending and tax cuts both expanded significantly, funded substantially by transaction-based taxes (stamp duty, VAT on construction) tied directly to an overheating property market. This left the exchequer's revenue base structurally exposed to a property market correction.

Monetary policy: a single ECB rate that was a poor fit for an overheating Ireland. As part of the eurozone, Ireland faced the same ECB policy rate as much larger, more sluggish economies like Germany and France. Those rates were arguably too low for an Irish economy running well above its sustainable trend — a textbook “one size fits none” problem of monetary union, since the single rate could not simultaneously suit an overheating Ireland and a stagnating German economy.

2008–2013: Financial Crisis, Bank Guarantee, and the Bailout

A blanket bank guarantee and severe fiscal consolidation. Following the property crash and global financial crisis, the government issued a blanket guarantee of the banking sector's liabilities in September 2008, and subsequently pumped tens of billions of euros into recapitalising and nationalising banks (including Anglo Irish Bank), pushing the sovereign's own debt burden to unsustainable levels. Ireland formally entered an €85 billion EU-IMF-ECB ("Troika") assistance programme in December 2010, exiting in December 2013 having implemented severe fiscal consolidation (spending cuts and tax rises, including a 14% cut to public sector pay in 2009) as a condition of the programme.

Evaluation: necessary, but arguably compounded a downturn that pro-cyclical boom-time policy had made worse. The scale of austerity required was directly a consequence of the fiscal position built up (and left exposed) during the pro-cyclical boom years — a clear illustration of why countercyclical fiscal discipline during a boom matters, since its absence left far less room to manoeuvre once the downturn hit. On the monetary side, Ireland had no independent lever to cushion the shock at all (no ability to devalue its own currency or set its own interest rate), leaving fiscal austerity as essentially the only adjustment tool available.

2014-2019: Recovery

A strong recovery, aided in part by ECB accommodation. Ireland's economy recovered strongly through this period, with unemployment falling from a peak of over 15% in 2011-2012 to under 6% by 2019. The ECB's own quantitative easing programme (asset purchases) from 2015 kept borrowing costs low across the eurozone, indirectly supporting Ireland's recovery and debt sustainability even though Ireland had no direct control over that policy.

2020-Present: COVID-19, and a Return of Fiscal Windfalls

A large, temporary fiscal expansion during COVID-19. The government ran significant deficits to fund the Pandemic Unemployment Payment, the Employment Wage Subsidy Scheme, and broader health and business supports — a genuinely countercyclical fiscal response to an exceptional shock, in contrast to the pro-cyclical pattern of the 2000s.

Since 2022-2023: a corporate tax windfall, and the return of an old lesson only partially learned. Ireland has recorded historically large budget surpluses since 2022, driven overwhelmingly by concentrated corporate tax receipts from a small number of large multinationals — corporate tax receipts reached €23.8 billion in 2023, and the 2024 surplus reached roughly 7% of modified GNI, boosted further by a €13 billion Apple back-tax payment following an EU Court of Justice ruling. To its credit, the government has explicitly avoided simply spending this windfall pro-cyclically, instead channelling it into two new sovereign wealth vehicles (the Future Ireland Fund and the Infrastructure, Climate and Nature Fund) targeting combined assets of over €100 billion by the mid-2030s — showing at least a partial, hard-won lesson learned from the concentrated, transaction-tax-dependent revenue base that left the exchequer so exposed in the 2000s. That said, the underlying revenue base remains heavily concentrated in a handful of large taxpayers, leaving genuine ongoing volatility risk the government itself openly acknowledges.

Monetary policy since 2022: another rate mismatch, in the opposite direction. The ECB's aggressive rate hikes from 2022 to fight eurozone-wide inflation were arguably a tighter setting than Ireland's own inflation and growth conditions alone would have called for, again illustrating the structural cost of ceding monetary policy entirely to a currency union average.

A strong answer makes the throughline explicit: Ireland's fiscal policy has swung between pro-cyclical (2000s boom, arguably worsening the eventual crash) and genuinely countercyclical (COVID-19 response, and more recent windfall saving via the sovereign wealth funds), while its monetary policy has been entirely determined by the ECB throughout, for better (crisis-era low rates, QE) and worse (a policy rate that has periodically been too loose or too tight for Ireland's own specific conditions) — the central, recurring critical point being that fiscal policy is the only lever Ireland itself actually controls.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Covers multiple distinct periods (Celtic Tiger boom, financial crisis/bailout, recovery, COVID-19, recent surplus) with specific facts for each, explicitly evaluates fiscal policy as pro-cyclical vs. countercyclical at different points, correctly notes Ireland's monetary policy is entirely determined by the ECB, and reaches a genuinely critical (not merely descriptive) overall assessment.
12-19	12-19	Covers two or three periods with reasonable factual detail but with limited critical evaluation, or doesn't clearly address the ECB/monetary policy dimension.
5-11	5-11	General, largely undated description of "austerity" or "recession" without specific periods, figures, or critical evaluation.
0-4	0-4	No meaningful attempt.

Question B2 Total: 100% (60% of the Paper)